

Migration Viability Score (MVS)

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The **Migration Viability Score (MVS)** is a structured decision-support model developed to quantify the comparative value of migrating from a legacy database system to the Oracle Database.

While not an established industry standard, MVS is grounded in multi-criteria decision analysis and is conceptually aligned with techniques used in enterprise architecture, IT governance, and financial justification frameworks.

Formula Definition

$$\text{MVS} = w_C \times \Delta C + w_P \times \Delta P + w_T \times \Delta T$$

Where:

- ΔC : Net cost differential (positive if migration reduces cost)
- ΔP : Personnel efficiency gain (positive if fewer specialized staff are needed)
- ΔT : Improvement in technical capabilities of the target system
- w_C, w_P, w_T : Assigned weights to reflect the relative importance of cost, personnel, and technical drivers respectively (where weights sum to 1)

Component Definitions

1. Cost Differential (ΔC)

$$\Delta C = C_current - C_migrated$$

Where:

- $C_current$: Total annualized cost of the current environment (including licenses, infrastructure, and support)
- $C_migrated$: Total projected cost of the target Oracle environment

2. Personnel Efficiency Gain (ΔP)

$$\Delta P = (N_current - N_migrated) \times S$$

Where:

- $N_current$: Number of full-time equivalents (FTEs) supporting the legacy environment
- $N_migrated$: Number of FTEs required after migration

- S: Average fully burdened annual salary per FTE

3. Technical Capability Improvement (ΔT)

$$\Delta T = T_{\text{migrated}} - T_{\text{current}}$$

Where:

T = Number of Required Capabilities Satisfied / Total Number of Required Capabilities

Capabilities may include:

- ACID transaction support
- SQL-based analytical capabilities
- Native JSON data handling
- Vector search and retrieval-augmented generation (RAG) compatibility
- Graph database features
- Geospatial support (e.g., OGC standards)
- MongoDB API emulation

Migration Decision Rule

- If $MVS > 0 \rightarrow$ Migration is Justified
- If $MVS \leq 0 \rightarrow$ Retain Current System

Example Calculation

Assume the following inputs:

- $w_C = 0.4$, $w_P = 0.3$, $w_T = 0.3$
- $\Delta C = 50,000$ USD (net annual savings)
- $\Delta P = 150,000$ USD (reduced labor costs)
- $\Delta T = 0.8 - 0.6 = 0.2$

The resulting Migration Viability Score is:

$$MVS = (0.4 \times 50,000) + (0.3 \times 150,000) + (0.3 \times 0.2) = 20,000 + 45,000 + 0.06 = 65,000.06$$

Conclusion: Based on the computed score, the migration is considered viable.

Notes

- The MVS model is adaptable and may be refined to incorporate additional drivers or adjusted weights based on organizational objectives.
- This model provides a transparent and repeatable method for comparing competing platform strategies and facilitates alignment across technical and business stakeholders.
- Questions/comments, please contact the author Matt DeMarco